

DISTRIBUTIONS OF STOPPING TIMES FOR SEQUENTIAL ELIMINATION RANKING PROCEDURES

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SUMMARY

Swanepoel (1977) has developed classes of sequential ranking and selection procedures which have the elimination property and which asymptotically satisfy certain probability requirements. Under a variety of circumstances, this note obtains the asymptotic distributions of his stopping times. In the indifference zone formulation, it is shown that his procedures are more efficient than those appearing in the literature except possibly in circumstances closely related to the least favourable configuration.

1. Introduction

Swanepoel (1977) has developed classes of sequential ranking and selection procedures which have the elimination property. This note studies the asymptotic distributions of a class of stopping times which include his procedures. We find a somewhat surprising deficiency in his original procedure, but a slight modification results in stopping times which are more efficient than those previously appearing in the literature.

Elimination rules are of obvious importance in the problem of selecting the best of k populations because they allow the experimenter the option of removing obviously inferior populations (compare the non-elimination procedures in Geertsema (1972) and Bechhofer, Kiefer and Sobel (1968)). Swanepoel's paper is noteworthy in that it is one of the first papers to adequately treat this topic, the most practical one in sequential ranking theory. However, he does not completely examine the number of observations the user can expect to save by using the elimination feature (i.e. the distributions of the stopping times), and it is to this problem that the current paper is addressed.

We deal only with the case of selecting the better of two populations for two reasons. First, the extension to three or more populations is straightforward (one of the advantages of studying elimination procedures) but notationally cumbersome. Second,

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the savings available from an early elimination of the inferior population when $k = 2$ will be illustrative and quite striking. This is especially in comparison to the Robbins, Sobel and Starr (1968) and Geertsema (1972) procedures which continue to take observations even when the decision is obvious.

In the general problem, we have k populations $\pi_1, \dots, \pi_k$ and a sample $X_{i1}, X_{i2}, \dots$ available from π_i . One forms location-scale equivariant statistics T_{in} from π_i based on $X_{i1}, \dots, X_{in}$ (i.e. $T_{in}(\xi(X_{i1}-\theta), \dots, \xi(X_{in}-\theta)) = \xi(T_{in}-\theta)$) and location invariant, scale equivariant variance estimates σ_{in}^2 . The assumptions are the existence of constants $\mu_1, \dots, \mu_k, \sigma_1^2, \dots, \sigma_k^2$ such that

$$(1.1a) \quad T_{in} - \mu_i = n^{-1} \sum_{j=1}^n Y_{ij} + R_{in}, \text{ where } Y_{i1}, \dots, Y_{in} \text{ are i.i.d. mean zero,} \\ \text{variance } \sigma_i^2 \text{ random variables with } E|Y_{i1}^3| < \infty$$

$$(1.1b) \quad n^{1/2} R_{in} \rightarrow 0 \text{ (a.s.)}$$

$$(1.1c) \quad \sigma_{in}^2 = \sigma_i^2 + o((\log_2 n)^{-1/2}) \text{ (a.s.)}$$

Assume without loss of generality that $\mu_1 \leq \dots \leq \mu_k$ and that it is desired to select the population with the largest value of μ_i . The indifference zone approach (we consider subset selection in the last section) assumes $\mu_k - \mu_{k-1} \geq \Delta$ and defines $r = r(\Delta) \rightarrow \infty$ as $\Delta \rightarrow 0$ (Swanepoel chooses $r = [1/\Delta]$, where $[\cdot]$ is the greatest integer function.) Let g be a given nonnegative function. Then Swanepoel's procedure eliminates population π_i at the n -th stage if there is a population π_j still in contention at the n -th stage for which

$$T_{jn} - T_{in} \geq r^{1/2} g(n/r) (\sigma_{in}^2 + \sigma_{jn}^2)^{1/2} / n - \Delta.$$

Sampling continues until only one population is left. If $W(t)$ is the standard Wiener process, and CS denotes a correct selection, Swanepoel shows

$$(1.2) \quad \liminf \Pr \{CS\} \geq 1 - \Pr \{W(t) \geq g(t) \text{ for some } t > 0\} (k-1).$$

If $k = 2$, the stopping time N_1 becomes

$$N_1 = \inf_{n \geq 1} \left\{ \begin{array}{l} T_{2n} - T_{1n} - (\mu_2 - \mu_1) \geq r^{1/2} g(n/r) (\sigma_{1n}^2 + \sigma_{2n}^2)^{1/2} / n - \Delta - (\mu_2 - \mu_1) \\ \text{or} \\ T_{2n} - T_{1n} - (\mu_2 - \mu_1) \leq -r^{1/2} g(n/r) (\sigma_{1n}^2 + \sigma_{2n}^2)^{1/2} / n + \Delta - (\mu_2 - \mu_1) \end{array} \right\}$$

Now define $T_n = T_{2n} - T_{1n} - (\mu_2 - \mu_1)$, $\mu = \mu_2 - \mu_1 \geq \Delta$ and $\sigma_n^2 = \sigma_{1n}^2 + \sigma_{2n}^2$. Then

the stopping time takes the form

$$N = \inf \left\{ n \geq 1 : \begin{array}{c} T_n - r^{1/2} g(n/r) \sigma_n / n + (\mu + \Delta) \geq 0 \\ \text{or} \\ T_n + r^{1/2} g(n/r) \sigma_n / n + (\mu - \Delta) \leq 0 \end{array} \right\}$$

where it is assumed in this paper that

$$(1.3a) \quad T_n = n^{-1} \sum_{i=1}^n Y_i + R_n, \quad n^{1/2} R_n \rightarrow 0 \text{ (a.s.)}, \quad EY_1 = 0, \\ EY_1^2 = \sigma^2 \text{ and } E|Y_1^3| < \infty.$$

$$(1.3b) \quad \sigma_n^2 = \sigma^2 + o((\log_2 n)^{-1/2}) \text{ (a.s.)}, \quad \sigma^2 = \sigma_1^2 + \sigma_2^2.$$

In Section 2, we investigate the stopping time N as $\Delta \rightarrow 0$ ($r \rightarrow \infty$), relate this back to the selection problem in Section 3, and discuss subset selection in Section 4.

2. Asymptotic Distributions

To begin with we will write $Y_i = \psi(X_i)$ to emphasize the dependence on the original sample $X_1, X_2, \dots$. It is not surprising that since the $\text{Pr}(\text{CS})$ is related to the Wiener process W , the distribution of N is also related to that of W . The proof of Theorem 1 gets its inspiration from work of Lai (1975) and, while self-contained, depends heavily on the approximations of Swanepoel.

Theorem 1

Suppose that $g(0) > 0$ and that g is a nondecreasing continuous function for $t \geq 0$ satisfying

$$(2.1) \quad t^{-1/2} g(t) \uparrow \infty \text{ as } t \downarrow 0 \text{ and as } t \uparrow \infty.$$

Then, for $t > 0$, if $r^{1/2}(\mu + \Delta) \rightarrow \infty$, $\text{Pr}(N/r > t) \rightarrow 0$. If $r^{1/2}(\mu + \Delta) \rightarrow c_1$, $r^{1/2}(\mu - \Delta) \rightarrow c_0$ ($0 \leq c_0 \leq c_1 < \infty$),

$$\text{Pr}(N/r > t) \rightarrow H(t) = \Pr \left\{ \begin{array}{l} \max_{0 \leq s \leq t} (W(s) - g(s) + sc_1/\sigma) \leq 0 \text{ and} \\ \min_{0 \leq s \leq t} (W(s) + g(s) + sc_0/\sigma) \geq 0 \end{array} \right\}$$

Proof of Theorem 1

Let $a > 0$ be arbitrarily small.

$$\begin{aligned}
 \Pr(N/r > t) &= \Pr \left\{ \begin{array}{l} \max_{n \leq [rt]} (T_n - r^{1/2} g(n/r) \sigma_n / n + (\mu + \Delta)) \leq 0 \text{ and} \\ \min_{n \leq [rt]} (T_n + r^{1/2} g(n/r) \sigma_n / n + (\mu - \Delta)) \geq 0 \end{array} \right\} \\
 &= \Pr \left\{ \begin{array}{l} \max_{1/r \leq s \leq t} (T_{[rs]} - r^{1/2} g([rs]/r) \sigma_{[rs]} / [rs] + (\mu + \Delta)) \leq 0 \text{ and} \\ \min_{1/r \leq s \leq t} (T_{[rs]} + r^{1/2} g([rs]/r) \sigma_{[rs]} / [rs] + (\mu - \Delta)) \geq 0 \end{array} \right\} \\
 &\leq \Pr \left\{ \begin{array}{l} \max_{a \leq s \leq t} (T_{[rs]} - r^{1/2} g([rs]/r) \sigma_{[rs]} / [rs] + (\mu + \Delta)) \leq 0 \text{ and} \\ \min_{a \leq s \leq t} (T_{[rs]} + r^{1/2} g([rs]/r) \sigma_{[rs]} / [rs] + (\mu - \Delta)) \geq 0 \end{array} \right\}.
 \end{aligned}$$

Let $W_r(s) = \sigma^{-1} r^{-1/2} [rs] T_{[rs]}$. Then the last expression is

$$\Pr \left\{ \begin{array}{l} \max_{a \leq s \leq t} (W_r(s) - \sigma_{[rs]} g([rs]/r) / \sigma + [rs](\mu + \Delta) \sigma^{-1} r^{-1/2}) \leq 0 \text{ and} \\ \min_{a \leq s \leq t} (W_r(s) + \sigma_{[rs]} g([rs]/r) / \sigma + [rs](\mu - \Delta) \sigma^{-1} r^{-1/2}) \geq 0 \end{array} \right\}.$$

Since $W_r \Rightarrow W$ on $D[a, t]$, $\sigma_{[rs]} / \sigma \rightarrow 1$ (a.s.), and g is continuous, the Theorem is proved if $r^{1/2}(\mu + \Delta) \rightarrow \infty$, while if $r^{1/2}(\mu + \Delta) \rightarrow c_1$, $r^{1/2}(\mu - \Delta) \rightarrow c_0$, the last probability must converge for fixed $a > 0$ to

$$H(a, t) = \Pr \left\{ \begin{array}{l} \max_{a \leq s \leq t} (W(s) - g(s) + sc_1 / \sigma) \leq 0 \text{ and} \\ \min_{a \leq s \leq t} (W(s) + g(s) + sc_0 / \sigma) \geq 0 \end{array} \right\}.$$

Since $H(\cdot, t)$ is continuous in a , $\lim_{r \rightarrow \infty} \Pr \{N/r > t\} \leq H(0, t)$. Hence, it suffices to show that as $r \rightarrow \infty$ and $a \rightarrow 0$, if $r^{1/2}(\mu \pm \Delta)$ have finite limits,

$$\Pr \left\{ \begin{array}{l} \max_{n \leq [ra]} (T_n - r^{1/2} g(n/r) \sigma_n / n + (\mu + \Delta)) \leq 0 \text{ and} \\ \min_{n \leq [ra]} (T_n + r^{1/2} g(n/r) \sigma_n / n + (\mu - \Delta)) \geq 0 \end{array} \right\} \rightarrow 1.$$

We only show that

$$G(a) = \Pr(\max_{1 \leq n \leq ra} (T_n - r^{1/2} g(n/r) \sigma_n / n + (\mu + \Delta)) > 0) \rightarrow 0,$$

as the other case is much easier. Now, for all $\epsilon > 0$ there exists n_0 such that $\Pr(\sigma_n^2 < n_0 \text{ for all } n \geq 1) > 1 - \epsilon$. Taking $n_0 = 1$ without loss of generality, it thus suffices to show that

$$G(a) = \Pr(\max_{1 \leq n \leq ra} (\sigma_n^{-1} T_n - r^{1/2} g(n/r) / n + (\mu + \Delta)) > 0) \rightarrow 0.$$

Now

$$\begin{aligned} G(a) &\leq \Pr(\sigma_n^{-1} \sum_{j=1}^n \psi(X_j) - W(n) \geq (r^{1/2} g(n/r) - n(\mu + \Delta))/4 \text{ for some } 1 \leq n \leq ra) \\ &\quad + \Pr(\sigma_n^{-1} n R_n \geq (r^{1/2} g(n/r) - n(\mu + \Delta))/4 \text{ for some } 1 \leq n \leq ra) \\ &\quad + \Pr(W(n) \geq (r^{1/2} g(n/r) - n(\mu + \Delta))/2 \text{ for some } 1 \leq n \leq ra) \\ &= P_2 + P_3 + P_4 \text{ (say).} \end{aligned}$$

We can make the substitution $n(\Delta + \mu) \sim nc_1/r^{1/2}$ and note that for $1 \leq n \leq ra$

$$\begin{aligned} r^{1/2} g(n/r) - n(\mu + \Delta) &\sim r^{1/2} (g(n/r) - nc_1/r) \\ &\geq r^{1/2} (g(n/r) - ac_1). \end{aligned}$$

Since g is increasing and $g(0) > 0$, we then obtain

$$(2.2) \quad r^{1/2} g(n/r) - n(\mu + \Delta) \geq r^{1/2} g(n/r)/2$$

for a sufficiently small. Thus we can replace the left side of (2.2) by its right side in P_2 ,

P_3 and P_4 . The rest of the proof follows from Swanepoel's equations (8)-(12).

Corollary 1

Suppose that Theorem 1 holds and in addition, that

$$(2.3a) \quad g(s)s^{-1+\epsilon} \rightarrow 0 \text{ as } s \rightarrow \infty \text{ for some } 1/2 > \epsilon > 0.$$

$$(2.3b) \quad g(s)/(s \log \log s)^{-1/2} \rightarrow \infty \text{ as } s \rightarrow 0.$$

Here $\log$ is the natural logarithm. Then $H(t) \rightarrow 1$ as $t \rightarrow 0$ and, if $c_1 > 0$, $H(t) \rightarrow 0$ as $t \rightarrow \infty$. If (2.3b) holds when $s \rightarrow \infty$, then if $c_0 = c_1 = 0$,

$$\liminf_{t \rightarrow \infty} H(t) > 0.$$

The proof of Corollary 1 is a consequence of the Law of the Iterated Logarithm for the Wiener process (Breiman (1968), page 266). The conclusion is that $1 - H(t)$ is a distribution function only if $c_1 > 0$. This has important consequences in Swanepoel's procedure ($r = 1/\Delta$) when $\mu = \Delta$, consequences which are discussed in detail in Section 3.

While it is clear that the proof of Theorem 1 breaks down ($N/r \xrightarrow{P} 0$) if $r^{1/2}(\mu + \Delta) \rightarrow \infty$, it is not clear why this should be so. The key is that if $r^{1/2}\Delta = o(1)$ (true in all our applications), then $r^{1/2}(\mu + \Delta) \rightarrow \infty$ implies that $r^{1/2}(\mu - \Delta) \rightarrow \infty$ and one begins to make correct selections with probability one and the problem becomes that of a one-sided stopping time as opposed to the original two-sided problem.

Lemma 1

Suppose that g is as in Theorem 1 and that, in addition

$$\int_{\eta}^{\infty} t^{-3/2} g(t) \exp(-g^2(t)/8t) dt < \infty$$

for some positive constant η . If $r^{1/2}(\mu - \Delta) \rightarrow \infty$, then $\Pr(CS) \rightarrow 1$. If for some

$$\epsilon > 0, \quad r^{1/2-\epsilon}(\mu + \Delta) \rightarrow \infty, \text{ then } N(\mu + \Delta) / r^{1/2} \xrightarrow{P} \text{og}(0).$$

Proof of Lemma 1 (μ fixed)

It is clear that $N \rightarrow \infty$ (a.s.) implies $\Pr(CS) \rightarrow 1$. This means that $\Pr(N \neq M) \rightarrow 0$, where

$$M = \inf \{n : T_n \geq r^{1/2} g(n/r) \sigma_n / n - (\mu + \Delta)\}.$$

Since $T_n \rightarrow 0$ (a.s.), $\sigma_n \rightarrow \sigma$ (a.s.) and $M/r \xrightarrow{P} 0$ by Theorem 1, an analysis along the lines of Chow and Robbins (1965) yields $r^{1/2}g(M/r)\sigma_M/M(\mu+\Delta) \xrightarrow{P} 1$. Since $M/r \xrightarrow{P} 0$ and g is continuous, the proof is complete.

Proof of Lemma 1 ($\mu \neq 0$)

By inspection or by Swanepoel's Theorem 3.1,

$$\Pr(\text{Incorrect Selection}) \leq \Pr(-T_n \geq r^{1/2}g(n/r)\sigma_n/n + (\mu-\Delta) \text{ for some } n \geq 1)$$

Now, for all $\epsilon > 0$ there exists $n_0 > 0$ such that

$$\Pr(\sigma_n^2 \leq n_0 \text{ for all } n \geq 1) \geq 1 - \epsilon,$$

so to prove $\Pr(\text{CS}) \rightarrow 1$ it suffices to show

$$\Pr(-\sigma_n^{-1}T_n \geq r^{1/2}g(n/r)/n + (\mu-\Delta)/n_0 \text{ for some } n \geq 1) \rightarrow 0.$$

Setting $n_0 = 1$, from the argument of Theorem 1 we see that as $a_1 \rightarrow 0$, $a_2 \rightarrow \infty$ the above probability is

$$\Pr(-\sigma_n^{-1}T_n \geq r^{1/2}g(n/r)/n + (\mu-\Delta) \text{ for some } a_1 r \leq n \leq a_2 r + o(1)).$$

Fix a_1, a_2 so that the $o(1)$ term above is arbitrarily small. Then the probability becomes

$$\Pr(\sigma_{[rt]}^{-1}[rt]r^{-1/2}T_{[rt]} + g\left(\frac{[rt]}{r}\right) + \frac{(\mu-\Delta)[rt]}{r^{1/2}} \leq 0 \text{ for some } a_1 \leq r \leq a_2) \rightarrow 0,$$

(as $r \rightarrow \infty$),

since $r^{1/2}(\mu-\Delta) \rightarrow \infty$. Thus, $\Pr(\text{CS}) \rightarrow 1$ and as before it suffices to consider M . It will be helpful to denote the dependence on μ by $M(\mu)$. We first want to show that

$$\log_2 M(\mu)/(M(\mu)\mu^2) \xrightarrow{P} 0;$$

by the Law of Iterated Logarithm this would imply that $T_{M(\mu)/(\mu+\Delta)} \xrightarrow{P} 0$ and hence that

$$\frac{r^{1/2}g(M(\mu)/r)\sigma}{M(\mu)(\mu+\Delta)} \xrightarrow{P} 1.$$

Since $M(\mu)/r \xrightarrow{P} 0$, this argument would then yield $(\mu+\Delta)M(\mu)/r^{1/2} \xrightarrow{P} \sigma g(0)$. We have already

shown that $M(1)/r^{1/2} \xrightarrow{P} \sigma g(0)$ since $(1+\Delta)M(1)/r^{1/2} \xrightarrow{P} \sigma g(0)$. Define $\eta_1 = (\mu+\Delta)^{1/4-\epsilon_1/2}$, where $\epsilon_1 > 0$ is very small. $M(\eta_1) \geq M(1)$ so that

$$\Pr(2M(\eta_1)\eta_1^2 \geq \sigma g(0)r^{1/2}(\mu+\Delta)^{1/2-\epsilon_1}) \rightarrow 1.$$

Since $\log_2 M(\eta_1) \leq \log_2 r$, this gives

$$\frac{\log_2 M(\eta_1)}{M(\eta_1)\eta_1^2} \xrightarrow{P} 0.$$

Hence, $\eta_1 M(\eta_1)/r^{1/2} \xrightarrow{P} \sigma g(0)$. Now let $\eta_2 = (\mu+\Delta)^{1/4-\epsilon_2/2}$. Then $M(\eta_2) \geq M(\eta_1)$ so that

$$\Pr(2M(\eta_2)\eta_2^2 \geq \sigma g(0)r^{1/2}(\mu+\Delta)^{3/4-\epsilon_2+\epsilon_1/2}) \rightarrow 1.$$

One continues the process, picking $\eta_3 = (\mu+\Delta)^{3/4-\epsilon_3/2}$ and carefully choosing $\epsilon_1, \epsilon_2, \dots$

We note that for technical reasons the approach of Lemma 1 cannot handle the case $\mu + \Delta = r^{-1/2}(\log_2 r)$, and we can see no way to get around this. However, in obtaining the asymptotic distribution of N , only the result of Lemma 1 is necessary. Recall again that if $r^{1/2}(\mu-\Delta) \rightarrow \infty$, we are essentially dealing with a one-sided stopping time, and it was this

that enabled us to show $M(\mu)(\mu+\Delta)r^{-1/2} \xrightarrow{P} \sigma g(0)$. From now on, let us assume

$$(2.4) \quad n^{1/2}(\sigma_n^2 - \sigma^2)$$

has a limiting distribution F and is uniformly continuous in probability (Anscombe (1952)).

De Wet and Venter (1974) have shown that (2.4) holds if T_{in} is the trimmed mean, while Carroll (1977) has verified (2.4) for certain classes of M -estimates.

Theorem 2

Assume Lemma 1 holds and that as $x \rightarrow 0$, $x^{-1/2}(g(x)-g(0)) \rightarrow 0$. Further assume that ψ is bounded and that $n^{1/2}(\sigma_n^2 - \sigma_{n-1}^2) \rightarrow 0$ (a.s.). Defining $x_0^2(r, \mu) = r^{1/2} \sigma g(0)/(\mu+\Delta)$, we have

$$\{r^{1/2} \sigma g(0) - (\Delta+\mu)N\}/x_0(r, \mu) = N^{1/2}(T_N - (\mu+\Delta)/(\sigma_N^2 - \sigma^2))/2\sigma^2 + o_p(1).$$

Remark

The referee has kindly pointed out that Theorem 2 has the following important

special case. Let us suppose that Lemma 1 holds, i.e., $r^{1/2}(\mu - \Delta) \rightarrow \infty$ and $r^{1/2 - \epsilon}(\mu + \Delta) \rightarrow \infty$ for some $\epsilon > 0$. Then, (1.3a), (2.4) and Theorem 2 imply that

$$\{r^{1/2}\sigma g(0) - (\mu + \Delta)N\}/x_0(r, \mu) \xrightarrow{L} N(0, \sigma^2),$$

where $\xrightarrow{L}$ denotes convergence in distribution and $N(0, \sigma^2)$ is the normal distribution with mean zero and variance σ^2 .

Proof of Theorem 2

Recall that $(\mu + \Delta)M(\mu)/r^{1/2} \xrightarrow{P} \sigma g(0)$. Since $\Pr(CS) \rightarrow 1$, it again suffices to consider M . Then

$$\begin{aligned} M^{1/2}T_M &\geq (M/r)^{-1/2}g(M/r)\sigma_M - (\Delta + \mu)M^{1/2} \\ &= (M/r)^{-1/2}(g(M/r) - g(0))\sigma_M + (M/r)^{-1/2}g(0)(\sigma_M - \sigma) + (M/r)^{-1/2}g(0)\sigma - \\ &\quad - (\Delta + \mu)M^{1/2}. \end{aligned}$$

Now, $r/M \xrightarrow{P} \infty$ so that

$$M^{1/2}T_M - (r/M)^{1/2}g(0)(\sigma_M - \sigma) \geq (r^{1/2}g(0)\sigma - (\Delta + \mu)M)/M^{1/2} + o_P(1).$$

Now, from (2.4) and Anscombe (1952), $M^{1/2}(\sigma_M^2 - \sigma^2)$ and hence $M^{1/2}(\sigma_M - \sigma)$ have limiting distributions. Thus,

$$g(0)(\sigma_M - \sigma) \{ (r/M)^{1/2} - (\mu + \Delta)M^{1/2}/\sigma g(0) \} \xrightarrow{P} 0.$$

Similarly,

$$(\mu + \Delta)M^{1/2} \{ (\sigma_M - \sigma)/\sigma = (\sigma_M^2 - \sigma^2)/2\sigma^2 \} \xrightarrow{P} 0,$$

which shows that

$$(2.5) \quad M^{1/2}(T_M - (\mu + \Delta)(\sigma_M^2 - \sigma^2)/2\sigma^2) \geq (r^{1/2}g(0)\sigma - (\Delta + \mu)M)/x_0(r, \mu) + o_P(1).$$

For the opposite inequality, note that since g is nondecreasing

$$\begin{aligned} M^{1/2}T_{M-1} &\leq (Mr)^{1/2}g((M-1)/r)\sigma_{M-1}/(M-1) - (\Delta + \mu)M^{1/2} \\ &\leq (Mr)^{1/2}g(M/r)\sigma_{M-1}/(M-1) - (\Delta + \mu)M^{1/2}. \end{aligned}$$

Then, since ψ is bounded,

$$M^{1/2}(T_{M-1} - T_M) = M^{-1/2}T_{M-1} - M^{-1/2}\psi(X_M) + o_p(1) = o_p(1).$$

By assumption, $M^{1/2}(\sigma_M^2 - \sigma_{M-1}^2) = o_p(1)$, so that the opposite to (2.5) is obtained, completing the proof.

To summarize this section, we have found the asymptotic distribution of N for the two population indifference zone selection problem in the two cases $r^{1/2}(\mu_2 - \mu_1 + \Delta) \rightarrow c_1 > 0$ and $r^{1/2 - \epsilon}(\mu_2 - \mu_1 + \Delta) \rightarrow \infty$. These turn out to be mathematically different because the latter case reduces to a one-sided problem. If $r^{1/2}(\mu_2 - \mu_1 + \Delta) \rightarrow 0$, partial results have been obtained.

3. Asymptotic Comparisons

The procedure suggested by Geertsema (1972), which like that of Swanepoel is essentially nonparametric but does not possess the elimination feature and assumes $\sigma_1^2 = \sigma_2^2 = \sigma_0^2$, leads to a stopping time which for $k = 2$ reduces to

$$N^* = \inf \{n \geq 5: n \geq \beta(P^*)^2(\sigma_{1n}^2 + \sigma_{2n}^2)/2\Delta^2\},$$

where $\beta(P^*)$ is defined for general k by

$$\int \Phi^{k-1}(x + \beta(P^*))d\Phi(x) = P^*,$$

and if $k = 2$, $\beta(P^*)^2 \sim -4 \log(1 - P^*)$ as $P^* \rightarrow 1$. One shows that, independent of $\mu_2 - \mu_1 = \mu$,

$$\Delta^2 N^* / \beta(P^*)^2 \xrightarrow{P} \sigma_0^2.$$

For the stopping time N defined in Section 1, Swanepoel chooses $r = 1/\Delta$ and defines

$$g(t) = (t + 1/4)^{1/2} \{\log(t + 1/4) - 2 \log(1 - P^*)\}^{1/2}.$$

He has shown that for $\mu_2 - \mu_1 = \mu$ fixed, as $P^* \rightarrow 1$,

$$(3.2) \quad EN^*/EN \rightarrow (1 + \mu/\Delta)^2 \quad (= 4 \text{ if } \mu = \Delta).$$

On the other hand, fixing P^* and letting $r^{1/2}(\mu_2 - \mu_1) = r^{1/2}\mu \rightarrow d_0$ ($0 < d_0 < \infty$), Theorem 1 shows that

$$N^*/N \xrightarrow{P} \infty.$$

The case not handled is the so-called "least favourable configuration"

$$\mu_2 - \mu_1 = \mu = \Delta = r^{-1},$$

but some insight may be gained by the following arguments. Fixing P^* , Theorem 1 shows that

$$(3.3) \Pr(N/r > t) \rightarrow c(t), \lim_{t \rightarrow \infty} c(t) = c > 0.$$

Now, suppose there exists $0 < \epsilon_1 \leq \epsilon_2$ with $r^{1+\epsilon_1} < N < r^{1+\epsilon_2}$ (a not unreasonable circumstance in light of (3.2) and (3.3)). Then, upon stopping, either

$$(3.4) \{N^{1/2}T_N + 2(N/r^2)^{1/2}\} / \{\log_2 N\}^{1/2} \geq (r/N)^{1/2} g(N/r) \sigma_N / \{\log_2 N\}^{1/2}, \text{ or}$$

$$(3.5) N^{1/2}T_N / \{\log_2 N\}^{1/2} \leq - (r/N)^{1/2} g(N/r) \sigma_N / \{\log_2 N\}^{1/2}.$$

Now, under these conditions for fixed P^* and by the Law of the Iterated Logarithm,

$$(r/N)^{1/2} g(N/r) \sigma_N / \{\log_2 N\}^{1/2} \geq \epsilon_1 \log r / \left\{ \log \{(1 + \epsilon_2) \log r\} \right\}^{1/2} \sigma_N.$$

$$|N^{1/2}T_N / \{\log_2 N\}^{1/2}| = O(1) \text{ (a.s.)}.$$

This means that $N/r^2 = \Delta^2 N = O(1)$ is impossible. Recalling that $\Delta^2 N^* = O(1)$, the preceding heuristic argument implies that if Δ is small relative to P^* and one is near a least favourable configuration, the Swanepoel stopping time can be worse than the Geertsema

proposal, i.e. it can happen that $N^*/N \rightarrow 0$.

Now suppose, however, that we define $r = ([1/\Delta])^2$. Then, since $\mu_2 - \mu_1 \geq \Delta$ the case $r^{1/2}(\mu_2 - \mu_1) \rightarrow 0$ is impossible and there are only two important cases

$$(i) \quad r^{1/2}(\mu_2 - \mu_1) \rightarrow d_0 \quad (1 \leq d_0 < \infty), \text{ i.e., } \mu_2 - \mu_1 \sim d_0 \Delta \text{ and}$$

$$(ii) \quad r^{1/2}(\mu_2 - \mu_1) \rightarrow \infty.$$

In the first case, Theorem 1 and Corollary 1 tell us that $N/r \sim \Delta^2 N$ has a limiting distri-

bution and, as $\Delta \rightarrow 0$,

$$\Pr(N^* < N) \sim \Pr(\Delta^2 N > \beta^2(P^*)\sigma_0^2)$$

$$\rightarrow \Pr \left\{ \begin{array}{l} \max_{0 \leq s \leq \beta^2(P^*)\sigma_0^2} (W(s) - g(s) + s(1+d_0)/(2\sigma_0^2)^{1/2} \leq 0 \text{ and} \\ \min_{0 \leq s \leq \beta^2(P^*)\sigma_0^2} (W(s) + g(s) + s(d_0-1)/(2\sigma_0^2)^{1/2} \geq 0 \end{array} \right\}$$

As $\beta(P^*)$ becomes large, in the least favourable configuration this approaches

$$\Pr \left\{ \begin{array}{l} \max_{0 \leq s \leq \infty} (W(s) - g(s) + s(2/\sigma_0^2)^{1/2} \leq 0 \text{ and} \\ \min_{0 \leq s \leq \infty} (W(s) + g(s)) \geq 0 \end{array} \right\} = 0,$$

the last following by the Law of the Iterated Logarithm. Thus, with $r = [1/\Delta]^2$ and even in the least favourable configuration, one cannot do much worse than the Geertsema procedure and frequently does much better. In case (ii) above, a simple analysis shows $N^*/N \xrightarrow{P} \infty$ as before. Similarly, for fixed μ , Δ as $P^* \rightarrow 1$, $T_N \rightarrow 0$ (a.s.) so that

$$r^{1/2} g(N/r) N^{-1} \rightarrow (\mu + \Delta)/(2\sigma_0^2)^{1/2}..$$

Taking g as above, this gives $N/\beta^2(P^*) \xrightarrow{P} \sigma_0^2/(\mu + \Delta)^2$ so that as $P^* \rightarrow 1$, $N^*/N \xrightarrow{P} 4$ in the least favourable configuration. We note in passing that for this choice of r , the correct selection probabilities are asymptotically bounded below by P^* .

4. Truncated Subset Selection

Swanepoel proposes a subset selection rule (see Gupta (1965)) in cases where at most r observations can be taken. Here $\Delta = 0$ and sampling proceeds as in the indifference zone problem with the exception that at the r -th stage sampling is discontinued and all remaining populations are declared "best". Swanepoel shows

$$\liminf_{r \rightarrow \infty} \Pr(CS) \geq 1 - \Pr(W(t) \geq g(t) \text{ for some } 0 < t < 1)(k-1).$$

Theorem 1 holds for $0 \leq t \leq 1$, while $\Pr(N > r) = 0$, and both Lemma 1 and Theorem 2

hold as stated.

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REFERENCES

- ANSCOMBE, J.F. (1952). Large sample theory of sequential estimation. *Proc. Camb. Phil. Soc.*, 48 600-617.
- BECHHOFFER, R.E., KIEFER, J. and SOBEL, M. (1968). *Sequential Identification and Ranking Procedures*. University of Chicago Press.
- BREIMAN, L. (1968). *Probability*. Addison-Wesley Publishing Company, Reading, Mass.
- CARROLL, R.J. (1977). On the asymptotic normality of stopping times based on robust estimators. To appear in *Sankhya, Series A*.
- CHOW, Y.S. and ROBBINS, H. (1965). On the asymptotic theory of fixed-width sequential confidence intervals for the mean. *Ann. Math. Statist.* 36 457-462.
- DE WET, T. and VENTER, J.H. (1974). An asymptotic representation of trimmed means with applications. *South African Statist. J.* 8 127-134.
- GEERTSEMA, J.C. (1972). Nonparametric sequential procedures for selecting the best of k populations. *Jour. Amer. Statist. Assoc.* 67 614-616.
- GUPTA, S.S. (1965). On some multiple decision rules. *Technometrics* 7 225-245.
- LAI, T.L. (1975). A note on first exit times with applications to sequential analysis. *Ann. Statist.* 3 999-1005.
- ROBBINS, H., SOBEL, M. and STARR, N. (1968). A sequential procedure for selecting the largest of k means. *Ann. Math. Statist.*, 39 88-92.
- SWANEPOEL, J.W.H. (1977). Nonparametric elimination selection procedures based on robust estimators. *South African Statist. J.* 11 27-41.